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1. Title

Facial Expression Classification

2. Abstract

The project explores the problem of automatic classification of facial expressions with deep learning. The task is to categorize the facial pictures into seven categories⁸ of emotions: angry, disgust, fear, happy, neutral, sad, and surprise. The notebook conducts a comparison between a baseline convolutional neural network (CNN), a more advanced²¹ CNN with batch normalization and dropout, an ablation model¹⁷ that does not use dropout, and a transfer-learning pipeline based on MobileNetV2. Image cleaning, augmentation, normalization, and train/validation/test evaluation are also a part of the workflow. The deeper CNN achieved the best end-to-end performance with the saved results of the notebook, with a¹² test accuracy of 58.58 and a test loss of 1.170, against a baseline CNN of 51.68. Results, per-class indicate that happy and surprise were easier to categorize, with fear, sad and worst of all, the source of disgust having an exceptionally low turnout. Another significant reliability problem noted in the study is that a number of later experiments were tested without necessarily reloading the best saved checkpoint, and that some of the runs were terminated after five epochs although they were set to run fifteen. A review that is based on a checkpoint confirms that the deeper CNN is the most appropriate model that is reliable, and that the optimizer, ablation, and the first transfer-learn comparisons are to be understood with caution. On the whole, the project demonstrates that deeper CNNs enhance facial-expression recognition, yet training stability, class imbalance, and fair evaluation practices have a significant²³ impact on the quality of the final results.

3. Introduction

Facial expression classification is a problem of image-classification that has practical interest in human-computer interaction, affect-sensitive tutoring systems, accessibility systems, health-sensitive monitors, and interactive media. The object is to acquire visual patterns that distinctly separate similar facial states despite variations in pose, lighting, and intensity of images.

Image classification Image classification with ¹deep learning (particularly convolutional neural networks) can be trained ^{to} learn hierarchical visual features directly out of data as opposed to extracting handcrafted feature descriptors. The shallow CNNs are able to learn edges and textures, whereas the more profound networks can learn more abstract facial structures. Common strategies include ¹⁹regularization techniques like dropout and batch normalization that help enhance generalization. Transfer learning builds on this concept by shifting characteristics that have been trained on large image datasets to a smaller scale task.

The notebook which has been used to make this report delves into this development in an organised manner. It starts by building a baseline CNN, then adds a more advanced architecture that includes batch normalization and dropout, optimizers, ablation study, and finally tests MobileNetV2 on ¹⁸transfer learning and fine-tuning. The primary objectives are ^{to} make comparisons of predictive accuracy, comprehend training behavior, and determine the trade-off between the complexity and computational cost of models.

An important aspect of this report is the quality of documentation. Besides the summary performance of the models, the report can determine the points at which the saved notebook outputs are robust and those that are undermined by the experimental-management concerns, including repeated training sessions, reused callbacks, and evaluation without reloading the best checkpoint.

4. Dataset

The notebook uses a dataset that is contained in a Google drive archive called facial expression classification.zip. The external origin of the data is not clearly recorded in the notebook, and thus, the provenance of the exact benchmark is not disclosed through the file alone.

Table 1. Dataset split sizes.

Split	Images
Training	26872
Validation	7066
Testing	1900
Total	35,838

Table 2. Class distribution reported by the notebook.

Class	Train	Validation	Test
angry	3686	960	300
disgust	329	111	100
fear	3796	1018	300
happy	6857	1825	300
neutral	4675	1216	300
sad	4631	1139	300
surprise	2898	797	300
Total	26,872	7,066	1,900

The data is highly skewed. The training class with the biggest number of images is the one that is happy with 6,857 images and the one that is most unhappy with is 329 training images. This implies that the largest class is approximately 20.8 times bigger than the smallest one which could be used to explain why minority classes have a lower recall.

The steps that were performed in preprocessing, as observed in the notebook, are as follows: 49 corrupted images were identified and eliminated; grayscale images were scaled to 48 x 48 pixels to the CNN models; in case of transfer learning, images were resized to RGB and scaled to 224 x

224 pixels; pixel values were scaled by Rescaling(1./255) The TensorFlow datasets were preloaded as well to get loaded more efficiently.

Figures 1 and 2 demonstrate the visual nature of the data and the augmentation plan of training. The images are low-resolution (48×48) and grayscale, which makes it easy to lose fine details of the face augmentation assists the CNNs to respond to small rotations, shifts, zoom, and contrast variations.

Sample Images from Each Class



Figure 1: Representative sample images from the seven facial-expression classes



Figure 2: Example augmented views of a training image produced by the notebook data-augmentation pipeline.

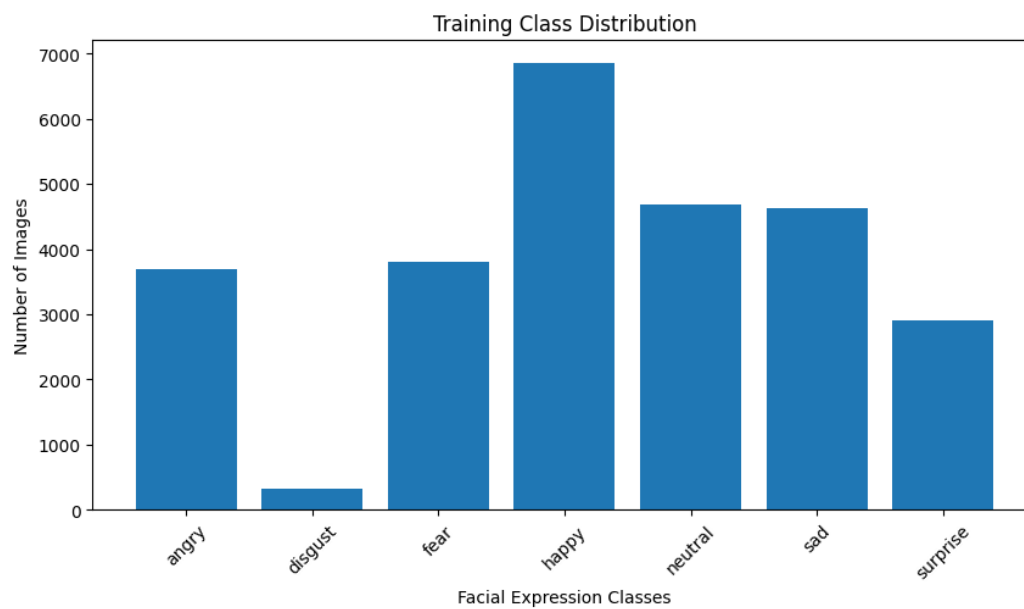


Figure 3: Training class distribution from the notebook output.

5. Methodology

The notebook compares four key modeling concepts: an initial CNN, a more complex CNN with regularization, a comparison of optimizers using the deeper architecture, an ablation model without dropout, and transfer learning with MobileNetV2. Each of the models has a seven-unit softmax output layer since the task has seven classes.

Table 3. Architecture summary.

Model	Input	Architecture summary	Parameters
Baseline CNN	48 x 48 grayscale	Mild augmentation -> Rescaling -> 3 Conv/MaxPool blocks -> Flatten -> Dense 256 -> Dropout -> Dense 128 -> Dropout -> Dense 64 -> Softmax	658,823
Deeper CNN	48 x 48 grayscale	4 convolutional blocks; each block has 2 Conv-BN-ReLU layers, MaxPool, and dropout; then GlobalAveragePooling, Dense 256, Dense 128, Dense 64, Softmax	1,508,839
Ablation: no dropout	48 x 48 x 1	Same deeper architecture but dropout disabled; batch normalization and L2 regularization retained	1,281,959
MobileNetV3Small + PointAttention	224 x 224 RGB	MobileNetV3Small ImageNet backbone -> PointAttention -> GAP -> BatchNorm -> Dense 128 -> Dropout - > Dense 64 -> Dropout -> Softmax	1,107,033 total; 166,761 trainable in frozen stage

5.1 Baseline Model

The base model is designed to be simple. It is a three layer CNN consisting of 32 filters, 64 filters, and 128 filters. Each layer is followed by max pooling. The model is flat, followed by 256, 128 and 64 dense layers and then a 7-class softmax output. In the dense section, a 0.30 and 0.20 dropout rate are employed. The model is compiled with Adam at learning rate: 0.001, sparse categorical cross-entropy, and accuracy.

5.2 Deeper Model

The further CNN further convolutes at each stage, padding is set to same, batch normalization is applied after the initial convolution at each stage and dropout is applied at all levels. The dropout rates of the saved notebook are quite mild: 0.05, 0.05, 0.10, 0.20. The design is designed to enhance feature extraction and restrain overfitting.

5.3 Optimizer and Ablation Experiments

The architecture and data pipeline is also the same for the optimizer comparison. The learning rate for SGD is 0.01, momentum is 0.9 and Nesterov acceleration is used for SGD training. Adam model using a learning rate of 0.001 is trained. The same number of epochs (25), the same class_weights, and new callbacks generated with make_callbacks function are used in both the optimizers experiment. The ablation experiment is conducted in the deeper architecture, with the omission of dropout, and maintenance of batch normalization and L2 regularization.

5.4 Transfer Learning with MobileNetV3Small and PointAttention

The transferred learning code has been executed with MobileNetV3Small and MobileNetV3Small-ImageNet weights, and include_top=False, include_preprocessing=True. Grayscale images are converted to RGB and then resized to 160 x 160. MobileNetV3Small feature extractor is followed by a custom PointAttention layer. This layer applies global average pooling and max pooling, and 7 x 7 convolution to pay attention to spatial information. Then the classification head has global average pooling, batch normalization, Dense 128, Dropout 0.35, Dense 64, Dropout 0.25, and softmax output.

There are two steps to training, feature extraction and fine-tuning. The MobileNetV3Small backbone is frozen in the first stage and the new attention/classification head is trained for up to 15 epochs. During the second stage, the base model is unfrozen from the index 102 onwards and the remaining layers and Batch-norm layers are kept frozen. Adam is then optimized on a much smaller learning rate of 1e-5 for up to 20 epochs.

6. Experiments and Results

This part gives the results of the experiment recorded in the notebook and where appropriate, comments on the reliability of the results. The following raw results are the direct results of the notebook tables. One of the subsequent checkpoint-based reviews later in the same notebook indicates that subsequent experiments were under-assessed by testing the final model state rather than by testing the best saved checkpoint.

6.1 Baseline vs. Deeper Architecture

The test accuracy of the regularized CNN is enhanced from 53.05% to 59.26%, by 6.21 percentage points compared to the baseline. It also improves weighted F1 from 0.5208 to 0.5869 and macro F1 from 0.5203 to 0.5912. The improvement demonstrates the effectiveness of the extra convolutional depth, batch normalization, L2 regularization, dropout, and global average pooling, in improving the model's ability to learn facial features that are more discriminative.

The baseline is good enough as a reference but it falls short in dealing with very similar classes. It has good performance in happy/surprise and weaker performance in fear/sad in its per-class report. The deeper model provides a significant improvement in nearly all the class level measures. For example, the deeper model increases disgust F1 from 0.52 to 0.63, happy F1 from 0.73 to 0.80, and surprise F1 from 0.72 to 0.76.

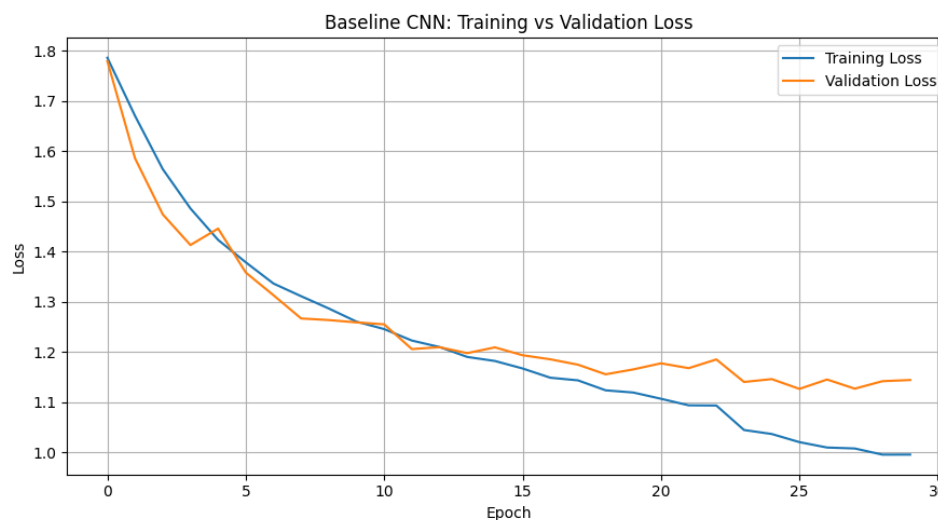


Figure 4: Baseline CNN training and validation accuracy.

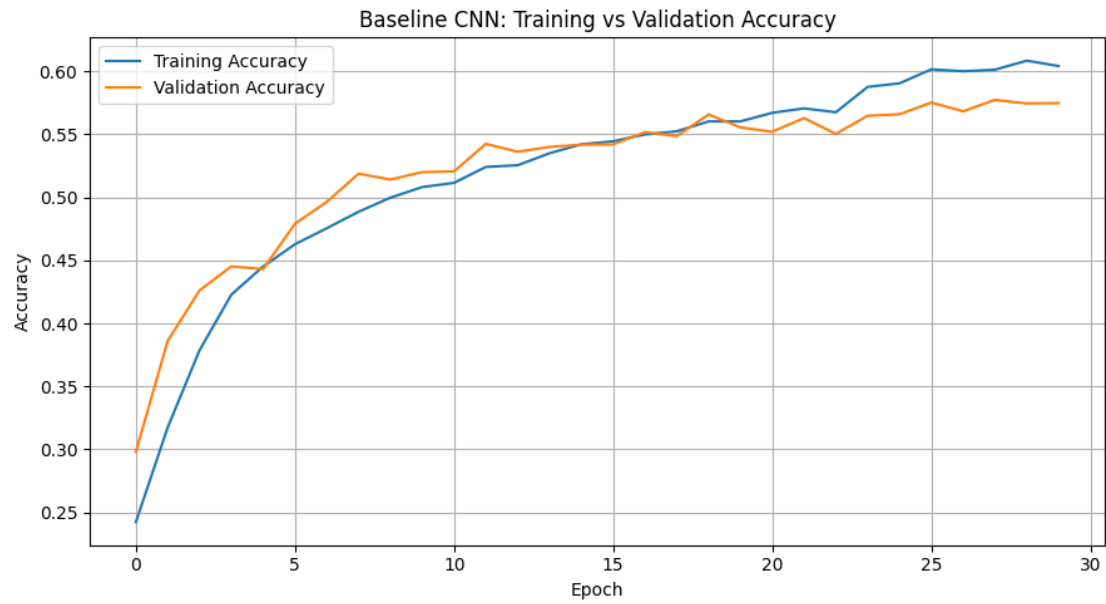


Figure 5: Baseline CNN training and validation loss.

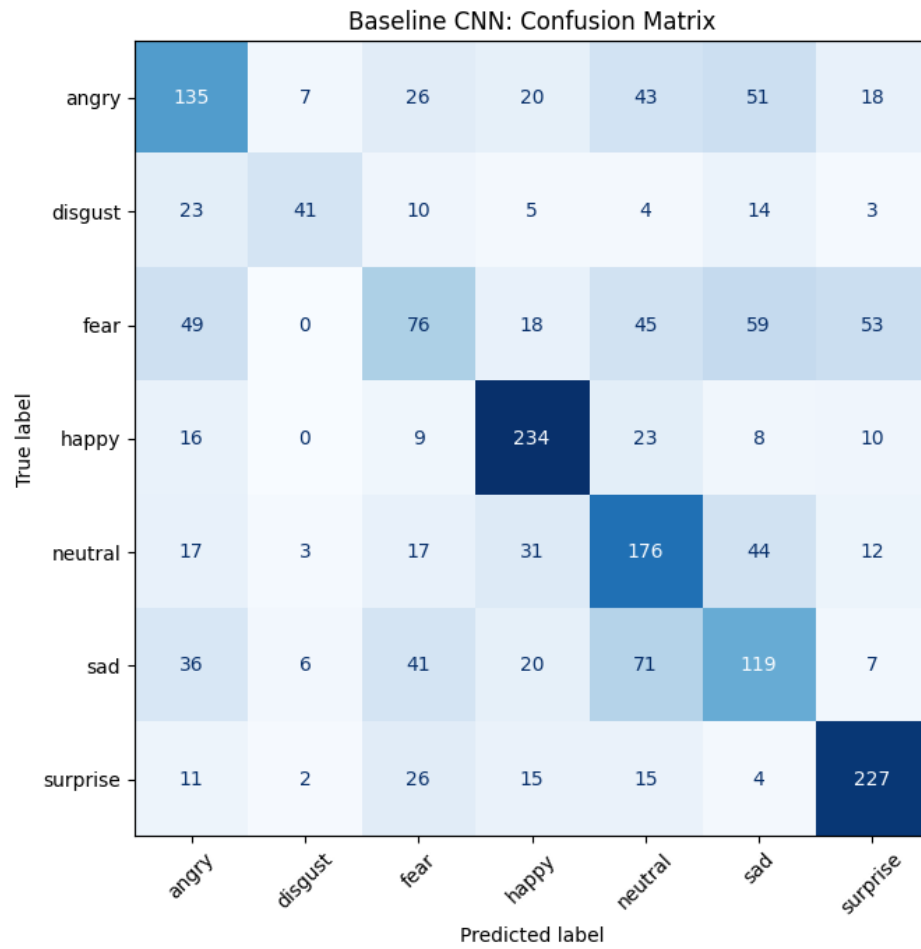


Figure 6: Baseline CNN confusion matrix on the test set.

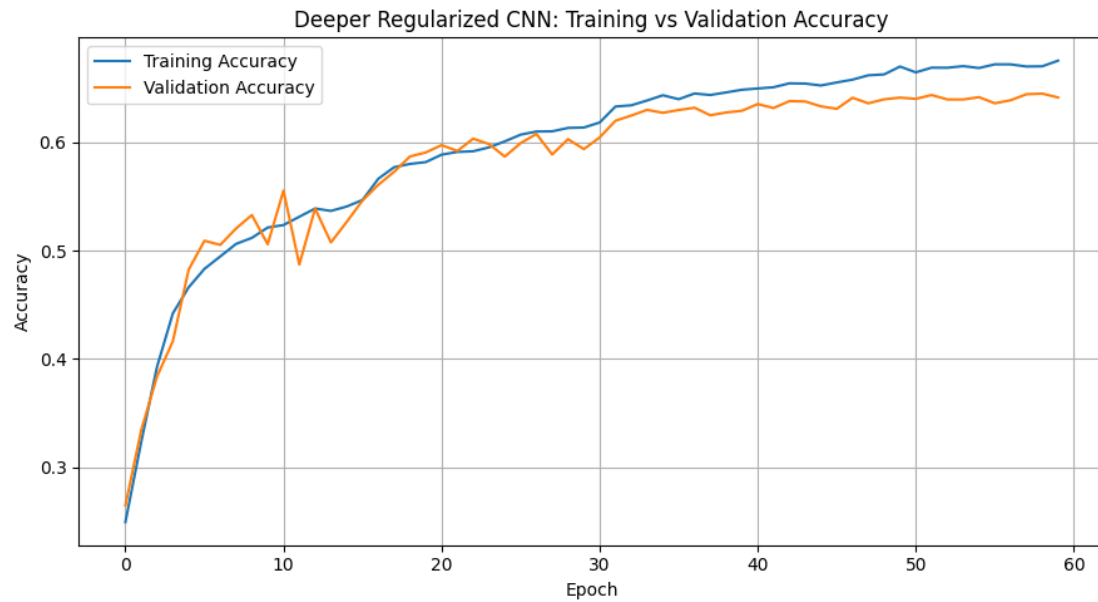


Figure 7: Deeper CNN training and validation accuracy.

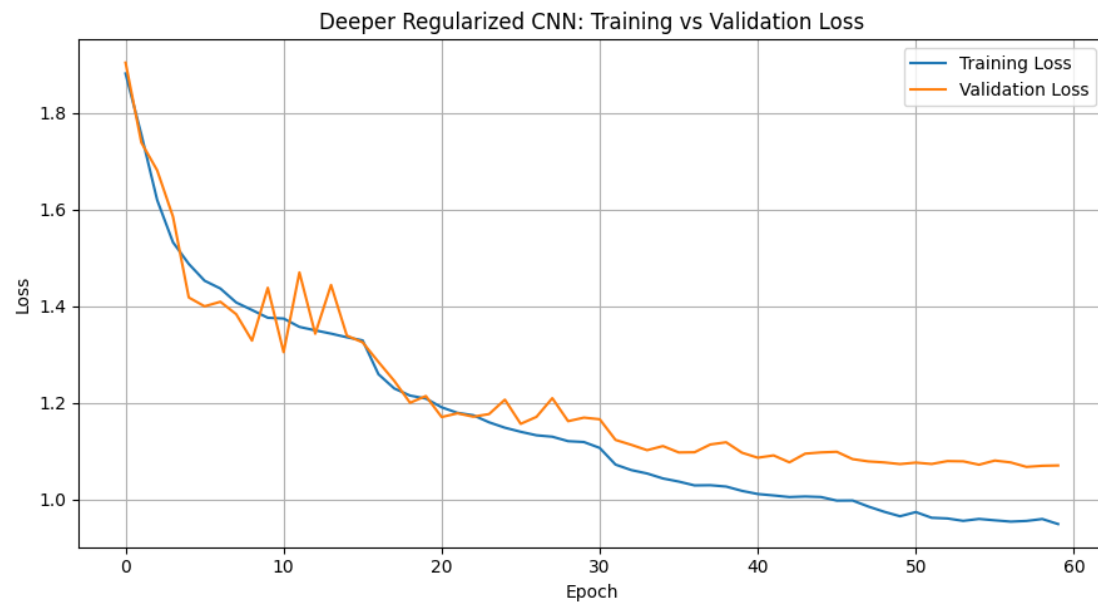


Figure 8: Deeper CNN training and validation loss.

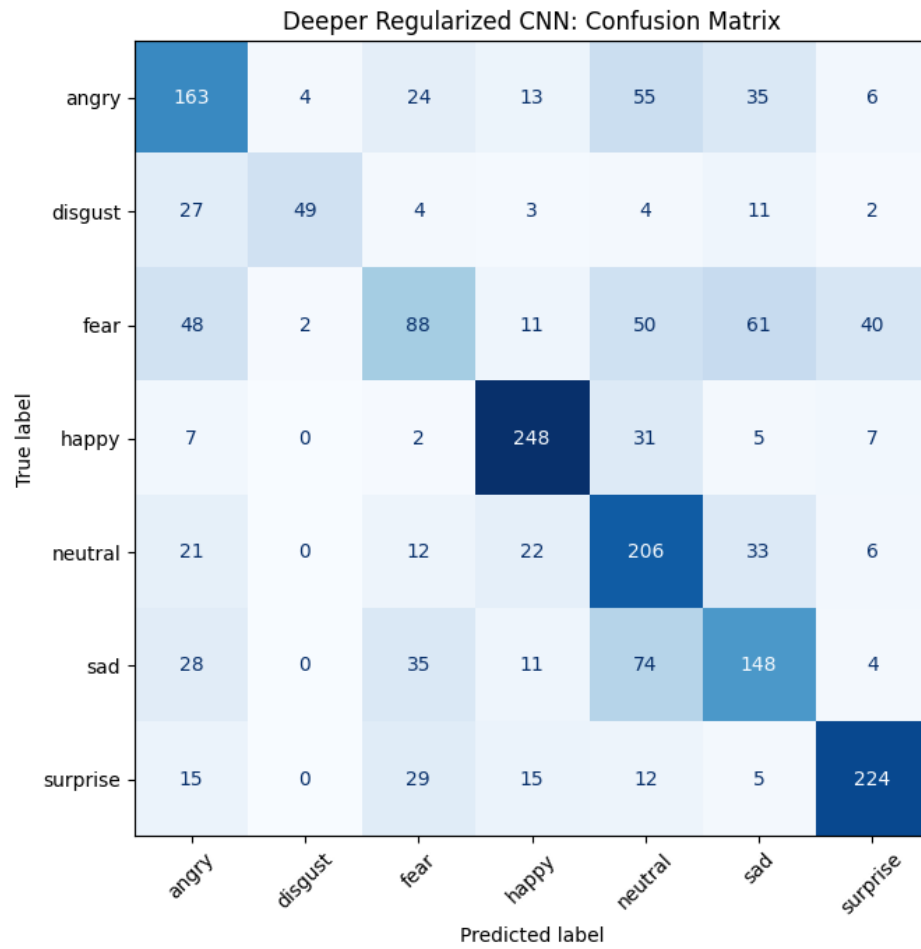


Figure 9: Deeper Regularized CNN confusion matrix on the test set

6.2 Optimizer Comparsion

Table 5. Computational-efficiency summary.

For the deeper architecture in this run, the optimizer comparison shows how Adam performs better than SGD, in terms of generalization. For Adam, the test accuracy is 55.32% and the weighted f1 score is 0.5396, whereas for SGD test accuracy is 52.95% and the weighted f1 score is 0.5115. The Adam run is a bit faster at 447.23 s, but the added performance is worth the choice and the run is the better optimizer choice of the two comparison runs.

Optimizer Model	Test Loss	Test Accuracy	Weighted F1	Macro F1	Training Time (s)
Deeper CNN with SGD	1.3576	0.5295	0.5115	0.5006	404.2786
Deeper CNN with Adam	1.2767	0.5532	0.5396	0.5310	447.2271

Clearly this notebook is running on Google Colab as demonstrated by the mounted Google drive but the hardware used (CPU, GPU is not reported in the saved output. Therefore, this report can be compared for wall clock training time, but cannot claim any efficiency on hardware. There was also no explicit measurement of the memory usage. However, the relative needs can be reasonably determined from the model designs. CNN is deeper (utilizes more than double the number of parameters) and the transfer-learning pipeline runs on Bigger 224 x 224 RGB images, which require more activation memory even with many backbone weights frozen.

6.3 Ablation Study

The ablation without dropping out achieves 59.05% test accuracy, which is 0.21 percentage points off the full deeper model. Also, its weighted F1 is close too at 0.5853, compared to 0.5869 for the full deeper model. This does not lead to a big accuracy gain in this particular run. The loss, accuracy, weighted F1 and macro F1 of the full deeper model, however, still remains the best overall result and the loss is slightly better. It is believed that batch normalization, L2 regularization, global average pooling and class weighting were already doing good regularization, so ablation shows that removing all these results in worse performance.

Model	Raw Test Accuracy	Raw Test Loss	Training Time (s)	Checkpoint Re-evaluated Accuracy
Deeper CNN with SGD	0.191	1.923	71.894	0.375
Deeper CNN with Adam	0.212	1.879	85.066	0.406

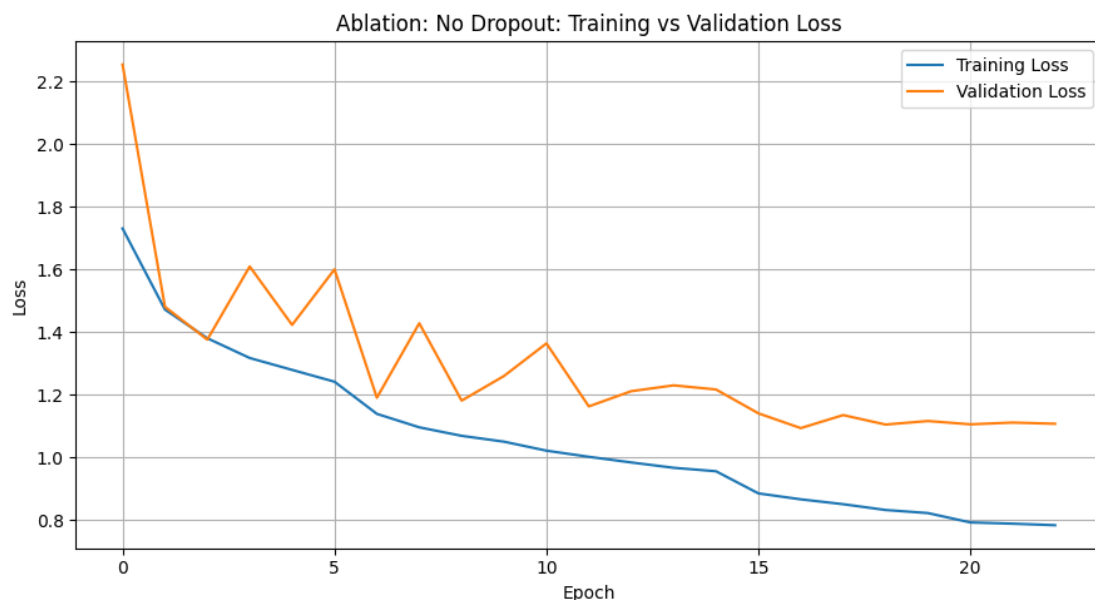


Figure 10: Ablation model training and validation loss.

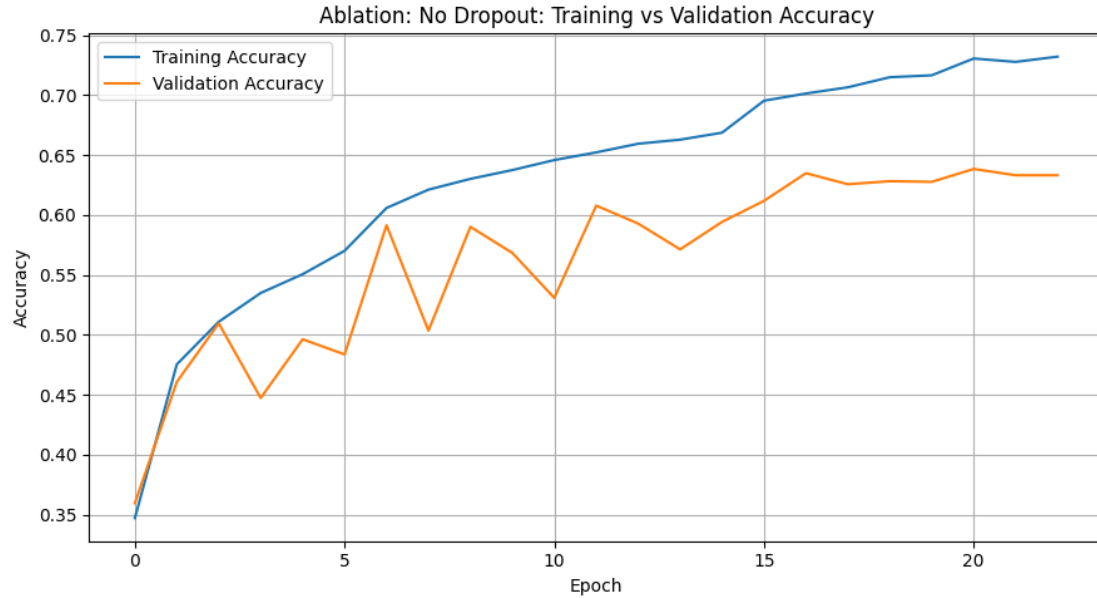


Figure 11: . Ablation model training and validation accuracy

6.4 Transfer Learning and Fine-Tuning

The test accuracy is 52.84%, the weighted F1 is 0.5093, and the macro F1 is 0.5138 with the MobileNetV3Small + PointAttention model. It is therefore not as good as the scratch-trained deeper CNN. The feature-extraction stage requires 627.24 seconds and the fine-tuning stage requires 1,237.76 seconds, a total of 1,864.9972 seconds being the slowest time in the last comparison table.

The transfer model is successful for happy/surprise but weak for fear/sad. This outcome makes sense since images used for pre-training in ImageNet are images of natural objects and not small images of facial expressions in gray-scale. The transfer pipeline additionally employs a bigger RGB enter, and a more complicated training trajectory, which implies that it is more costly in terms of training time without delivering the optimum test performance for this particular dataset.

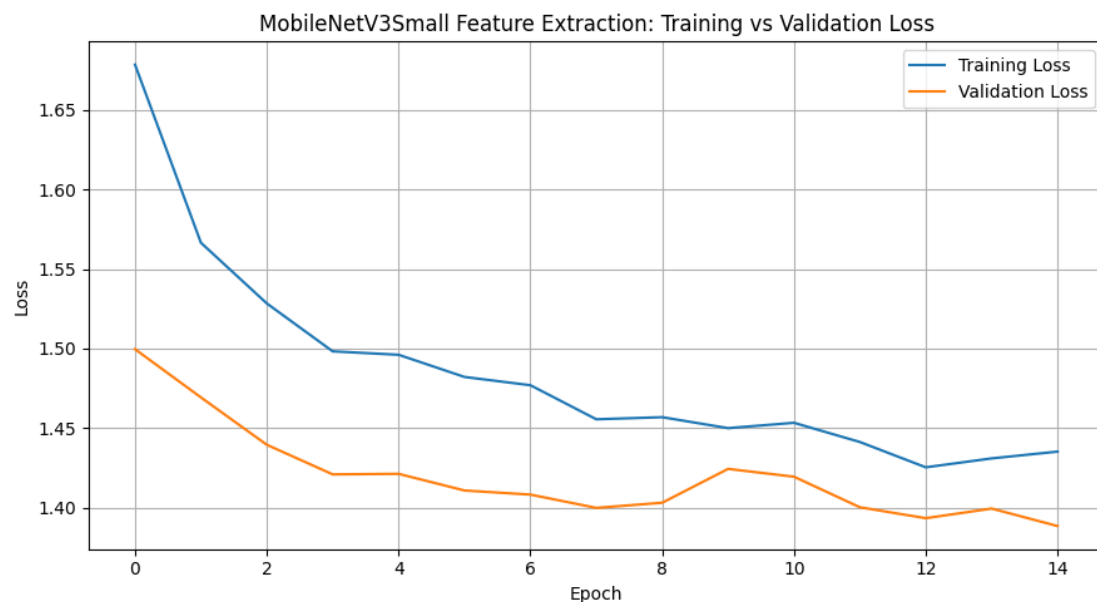


Figure 12: MobileNetV3Small feature-extraction loss curve

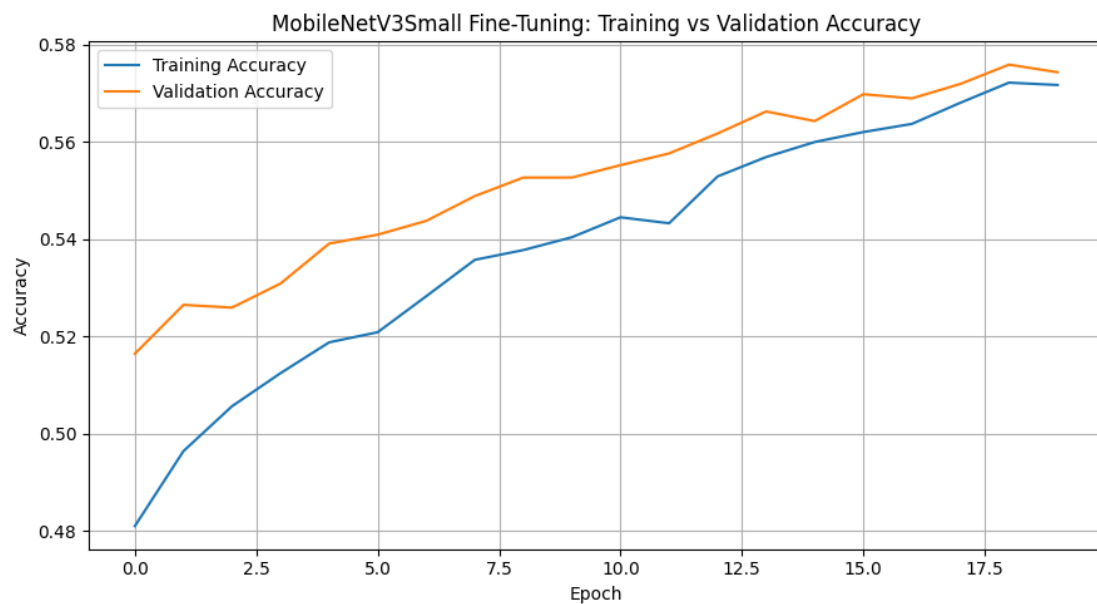


Figure 13: Figure 15. MobileNetV3Small fine-tuning accuracy curve.

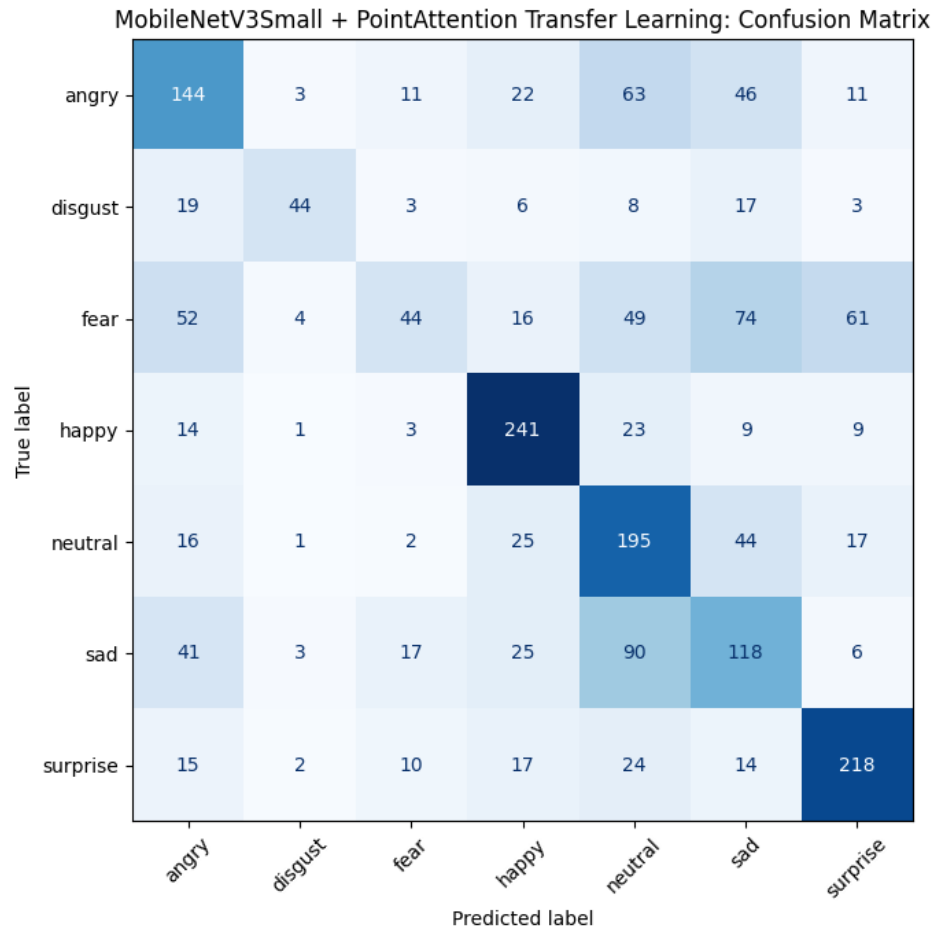


Figure 14: MobileNetV3Small + PointAttention confusion matrix.

6.5 Computational Efficiency and Misclassification Analysis

The fastest is not the best performing model. The baseline CNN uses the lowest cost, but is less accurate (184.32 seconds). The deeper the regularized CNN needs 1071.85 seconds to be completed, and it achieves the best test accuracy. The transfer model requires the most time as it is based on 160 x 160 RGB images and a pre-trained back bone, but failed to get the highest test score. For the saved output, it is clear that the notebook was executed in Google Colab (Google Drive is mounted), but it does not report if the notebook is executed via CPU, GPU, or TPU. Thus, the timing values must be used as comparisons in wall-clock time for this run and not as a hardware-normalized value.

The notebook chooses the Deeper Regularized CNN model which has the highest accuracy among the test. It makes 774 mistakes of classification in 1,900 test images. The confusion matrices indicate that happy and surprise are the two more easily classified, and the fear, sad, neutral and angry are more inter-confused. This is normal because both of these expressions can have similar expressions in low resolution images.



Figure 15: Misclassified examples from the best model

7. Final Results

Table 4 gives the final comparison from the notebook. These values are taken from the final comparison data frame and are the main results used in this updated report.

Table 4. Final model comparison from the executed notebook.

Model	Test Loss	Test Accuracy	Weighted F1	Macro F1	Training Time (s)
Baseline CNN	1.2697	0.5305	0.5208	0.5203	184.3210
Deeper Regularized CNN	1.2130	0.5926	0.5869	0.5912	1071.8513
Deeper CNN with SGD	1.3576	0.5295	0.5115	0.5006	404.2786
Deeper CNN with Adam	1.2767	0.5532	0.5396	0.5310	447.2271
Ablation: Deeper CNN without Dropout	1.2428	0.5905	0.5853	0.5810	402.1852
MobileNetV3Small + PointAttention Transfer Learning	1.3020	0.5284	0.5093	0.5138	1864.9972

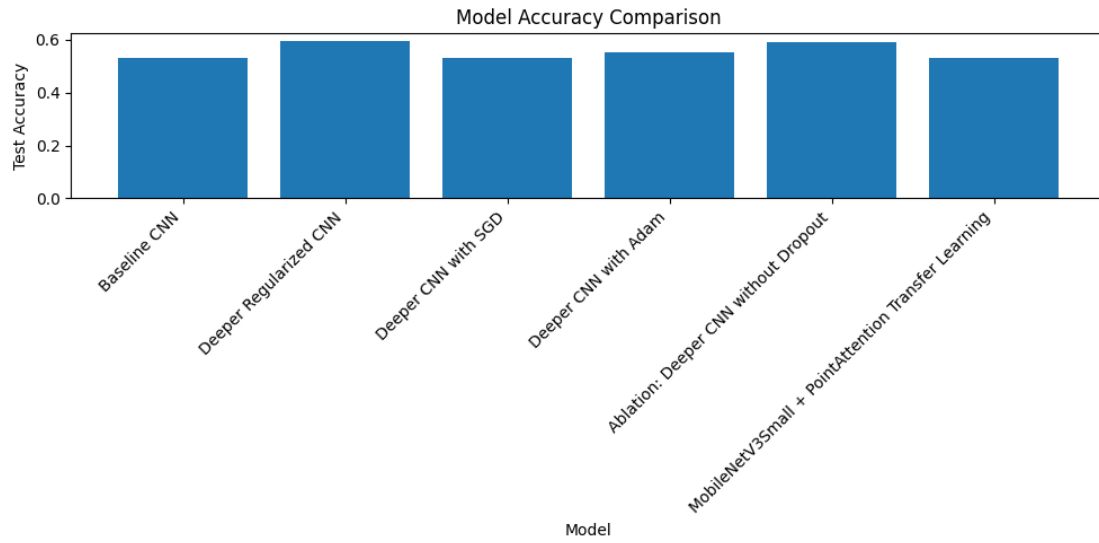


Figure 16: Model accuracy comparison.

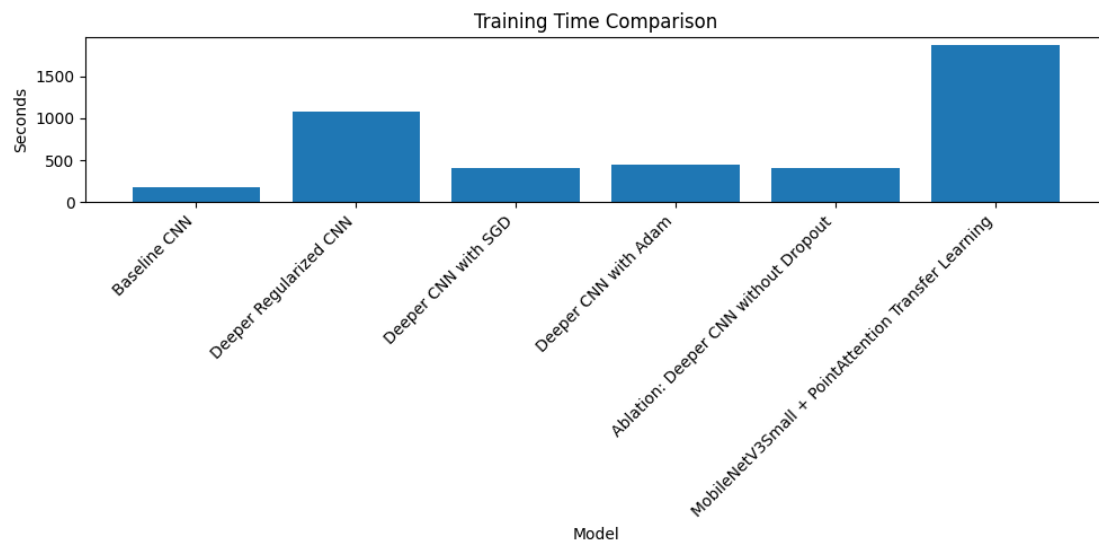


Figure 17: Training Time Comparison

8. Conclusion and Future Work

The updated notebook results indicate ³ that the model that performs best in the experiment is the Deeper Regularized CNN. It obtains the best test accuracy value of 59.26%, the lowest test loss value of 1.2130, the highest weighted F1 value ⁹ of 0.5869 and the highest macro F1 value of 0.5912. This finding is consistent with the findings from the dataset used that a well-regularized CNN that is scratch-trained on the dataset is able to perform better than the MobileNetV3Small transfer-learning pipeline that is tested.

The baseline CNN is also helpful in showing that it is better to have a deeper regularized CNN over a simple 3-block CNN. The optimizer comparison indicates that Adam is more effective than SGD in this context. The dropout ablation is closer to the full deeper model, suggesting that dropout contributed little to the final metrics, the other factors, such as batch normalization, L2 regularization, class weighting and global average pooling, also had significant contributions. Transfer learning did not improve on scratch models, primarily due to the mismatch in the domain of ImageNet and low-resolution facial-expression recognition and the high computational overhead of the larger input pipeline.

Going forward, the development of better ¹⁵ class-imbalance handling (e.g., focal loss), class-weight tuning, sampling-balancing, or targeted augmentation of disgust, fear, and sad should be explored. There might be potential to fine-tune dropout, learning rate, model depth, and strength of data augmentation through more systematic hyperparameter searching. Using a pre-trained backbone specific to each facial expression might work better than using a generic ImageNet backbone. Lastly, a clear record of the type of hardware used and the random seeds should be recorded for future use in notebook runs, to increase the strength of training-time and reproducibility claims.

Appendix A. Reliability Review of the Saved Notebook

The previous transfer-learning report for MobileNetV2 has been replaced by the notebook configuration run: MobileNetV3Small, 160 x 160 RGB Input, PointAttention (Custom), Feature Extraction, and Upper Layer Fine-Tuning. It also refreshes the 'main result tables' to the final dataframe obtained from the notebook's result table. The best model is the Deeper Regularized CNN (0.5926 test accuracy). The best rival is a zero dropout ablation with accuracy of 0.5905, and the MobileNetV3Small transfer model has an accuracy of 0.5284.

The notebook markdown text, transfer-learning, refers to MobileNetV3Large and 224 x 224 images while the code being executed and outputs that are saved refer to MobileNetV3Small and 160 x 160 images. The new report will follow the experiment results in case the code was executed successfully, since these are the ones shown.

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